



Module 1 Day 2: Attacking and Defending

Class Instruction

Overview

Today, you will assess attack and defense strategies of a vulnerable web login.

Class Objectives

By the end of class, you will be able to:

- Understand the Security Mindset.
 - List different types of user, web, server, and database cybersecurity attacks.
 - Identify risk mitigation plan frameworks for user, web, server, and database cybersecurity attacks.
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01. Welcome and Overview

- In the previous class, you were introduced to the structure of the course and the staff who will support you. We also covered the rise of cybersecurity, the boot camp's course overview, and the technical infrastructures we will use to complete labs. Finally, you were introduced to the fundamental cybersecurity concept of the CIA triad.

Cybersecurity focuses on two primary concepts:

- **Threat assessment:** The structured process of identifying the threats posed to a group or system.

- **Risk mitigation:** The systematic reduction of the impact and likely occurrence of a negative event.

Why do we use the word "mitigation" and not "eradication"?

- Answer: We cannot eradicate all risks. Since cybersecurity is an ever-evolving landscape, new threats emerge every day. Also, business needs and budgetary constraints often limit the ability to implement cybersecurity best practices. So, we must do our best to mitigate.

Today's Topics

Today's class will continue with assessing risk and mitigating threats by evaluating specific attacks and vulnerabilities of users, web applications, servers, and databases. In short, you will be thinking about ways to attack and defend pillars of organizations in order to access and protect valuable information.

In today's class, you will have the opportunity to think like offensive and defensive cybersecurity professionals evaluating the attacks and defenses of levels of information within a company.

- To assess threats and mitigate risks, you need to check each component of an organization, and understand how malicious actors can exploit weaknesses within the organization and damage the stakeholders' finances, reputations, and well-being.

02. Activity Security Task #1 - The Stolen iPhone

- [M1.2 Activity File: The Stolen iPhone](#)

03. Instructor Review: Security Task #1 - The Stolen iPhone

This activity had several purposes:

- To practice creative thinking.

- To understand that thinking like a cybercriminal can help cyber professionals determine mitigation methods.
 - To determine different types of potential vulnerabilities.
 - To demonstrate that these points all make up the Security Mindset.
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04. Introducing Security Task #2 - Attacking the Wall

You will now work on two security tasks related to assessing threats and mitigating risks.

In this next activity, you will continue to use the Security Mindset, but focus on technical systems that are common with many organizations.

- You will examine various attack strategies that cybercriminals can use to penetrate insecure logins.
- This exercise should encourage you to think creatively about all the ways to penetrate a system, from user attacks to physical break-ins.
- To successfully complete this exercise, think creatively. You will experience a real sense of accomplishment when you discover real-world attack types through discussion and brainstorming.

Using the slides, review the scenario.

- You and your team have just been hired by a very successful startup, Bitcoin Dating Exchange.
- While their founding team is brilliant, like many startups, they don't know the first thing about security.
- They have just handed you a lot of money to solve their most critical problem: their log-in process is totally insecure. Malicious attackers are routinely logging in as users (and administrators) and gaining access to company data and financial assets.

In groups, develop a list of 15 ways that a malicious actor could penetrate the system and log in as a user or administrator.

- With each method, be prepared to describe the following:
 - Who (or what) is the initial target?
 - How would the actor implement the attack?
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05. Activity - Security Task #2 - Attacking the Wall

- [M1.2 Activity File: Attacking the Wall](#)
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06. Instructor Review: Security Task #2 - Attacking the Wall

Now that we have evaluated all the threats posed to this company, we will consider how to mitigate the risks.

07. Break (0:15)

08. Introducing Security Task #3 - Defending the Wall

Before the break, your team discovered at least 10 different ways to attack a system. Now it's time to think about mitigation strategies. Remember: Be creative and think of both technical and non-technical solutions.

09. Activity - Security Task #3 - Defending the Wall

- [M1.2 Activity File: Defending the Wall](#)

In the same groups, you will now use the list of potential attacks from the previous activity to develop a list of at least 10 strategies to mitigate the website's risk.

- The goal of this exercise is for you to realize that you can use certain mitigation strategies to handle multiple threats at once.
 - You should arrive at a total of 10 or more strategies that address the concerns.
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10. Instructor Review: Security Task #3 - Defending the Wall